

OPERATIONS MANAGEMENT

What is Operations Management?

Operations management is the administration of business practices to create the highest degree of efficiency possible within an enterprise. It involves utilizing resources from people, materials, equipment, and technology. Operations management aims to keep critical quality standards while ensuring that day-to-day operations are efficient, cost-effective, and timely. It also aims to strategize how to effectively build structures for its operations, manipulate and utilize information technology networks to hasten work processes, and use proper inventory systems (Kenton, 2019).

Role of Operations Management in a Business

The responsibilities of operations management range from the following (Dhoul, 2022):

- **Capacity planning.** It involves evaluating the number of products or services a business can sell or distribute in a given time frame.
- **Product design and service design.** To ensure that customer requirements and expectations are met, new concepts are generated, and a service or product is developed. It also involves a product's cost-effectiveness and meeting customer requirements.
- **Quality control.** Quality management or assurance involves checking for potential problems or errors in services or products at every stage of the production process or during service operations.
- **Process improvement and optimization.** It involves evaluating the steps in a process, either completely rewriting it or rearranging the steps to work best.
- **Supply chain management.** It involves controlling the sourcing of supplies, the production process, inventory management, sales, and distribution at reasonable rates while managing the supply chain process. As a result, overhead costs are reduced, production is efficient, and products are delivered on time.

Location and Start-Up Needs

Facilities Management. Facilities Management is under the umbrella of operations management. It integrates people, places, and processes, as well as the handling and care of structures, buildings, equipment, and other physical necessities of a business so that the productivity of the operations will be above par.

As an enterprise grows, so will its assets. Businesses, even start-ups, tend to acquire assets to aid them in delivering goods and services to their customers. Assets are properties owned by an enterprise regarded as having a value that can be used to support the production of goods and services.

These assets could be machines, equipment, tools, and structures. These facilities need periodic inspections and maintenance to prevent them from interrupting company operations, which entail expensive repairs (Christiansen, 2019).

Facilities management is segregated into two (2) categories:

1. **Hard services** relate to any physical alterations and adjustments in the building or its structure. A common example of hard service is fire safety, which includes the installation of water sprinklers, fire alarm systems, fire extinguishers, closed-circuit television cameras, theft alarms, electrical and air-conditioning systems, and other mechanical services.
2. **Soft services** are responsible for making the workplace bearable and comfortable. These include landscaping, event support, cleaning services, canteens and pantry, pest control, and others that uplift employee morale.

Facility Location. Facility location involves learning a geographic site for an enterprise's operations. An enterprise must consider several factors when selecting a location site for operations. Examples are proximity to customers and suppliers, labor costs, employee rates, transportation, etc.

Characteristics of tangible and intangible factors under facility location:

- a. **Tangible factors** include delivery or freight rates, production costs, local government tax cuts, and production costs.
- b. **Non-tangible factors** include the location's security, quality of work, and culture.

The following are the things to consider when deciding where to set up the business (Wisdom IT Services, 2018):

A. Controllable factors

- **Proximity to the market** – Every business is expected to deliver its goods and services to its customers promptly. It can lead a business to situate itself close to its market.
- **Supply of materials** – The enterprise must acquire raw materials in the right qualities, quantities, and time to produce uninterrupted. This factor becomes very important if the materials are perishable and the cost of transportation is very high.
- **Transportation facilities** – Quick transport facilities ensure a timely supply of raw materials to the business and finished goods to the customers. The transport facility is a prerequisite for the location of the plant. It is composed of five (5) modes of transporting raw materials: via air, rail, water (seas and rivers), roads, and pipelines (water supply, gas supply, and fuel supply).
- **Infrastructure available** – Businesses dependent on energy, like steel mills, should be close to power stations for an uninterrupted power supply. Businesses that rely heavily on technology and internet support should be near communication companies.
- **Labor wages and the laws that govern it** – An enterprise should always be aware of its workers' rights and the tax laws where the business is located. Here in the Philippines, wages are often smaller in the provinces than in the cities due to the high cost of living. It is also true for taxes.

B. Non-controllable factors

- **Government policies** – These include the policies of the government and local bodies concerning labor laws, building codes, safety, taxes, social security, data privacy, and other factors that demand attention. Every enterprise should know these laws to avoid administrative and criminal liability.

- **Climate and environmental conditions** – The business must know its location's geology, climate, and environmental conditions. These factors can significantly affect the delivery of services and products to customers.
- **Supporting industries** – Businesses must have a good relationship with their suppliers. These suppliers can make or break an enterprise. Also, the proximity of the supplier is essential to ensure that the delivery of quality raw materials is done in a timely fashion, thus avoiding hassle on the part of the customer.

Workplace Layout

An office layout that was once a cubicle or a room enclosed by four walls has evolved into an open, airy space that makes collaborating easier and gives employees more visibility. Office layout plans define the architecture of collaboration, cross-functional operations, and elements that focus on maximizing productivity, learning, and employee satisfaction (Brady, 2021).

- A. **Open Office Design.** Workstations are arranged in various ways, some touching each other and others clustered in work zones of varying sizes. Managers and everyone in the office sit together, allowing for open discussion throughout the open area. Additionally, it may result in operational efficiencies. Workers' ability to communicate directly in open offices instead of relying on e-mail has been credited with significant reductions in e-mail traffic and significant increases in decision-making.

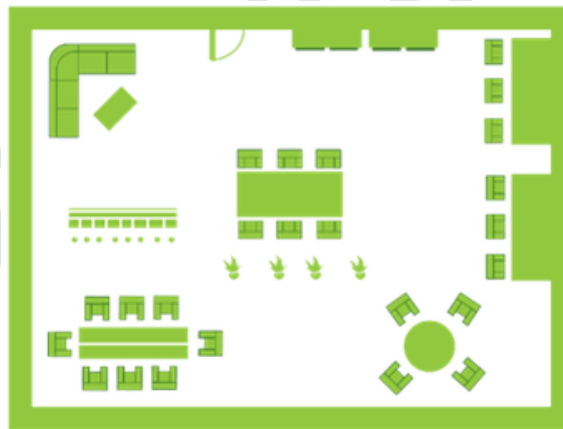


Figure 1: Open Office Design

Source: <https://www.newdayoffice.com/blog/how-to-choose-the-best-office-layout-for-your-business>

- B. **Cellular Office Design.** The entire office space is divided into individual spaces or cubicles, so employees can each have private areas in this popular layout. It facilitates the systematization of office operations and provides a focused work environment. Individual spaces are made by using partitions, windows, doors, or studs in the ceiling.

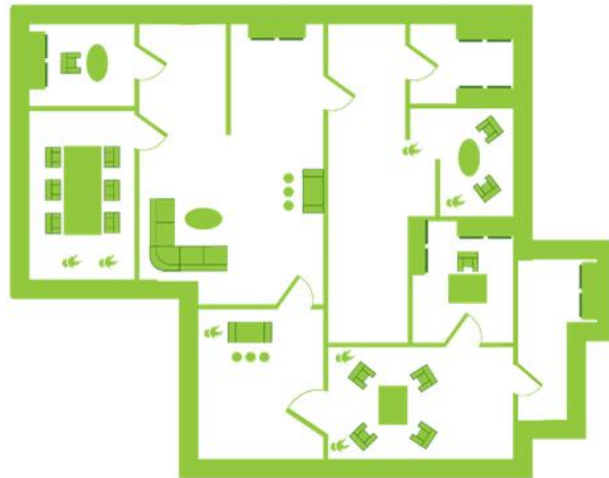


Figure 2: Cellular Office Design

Source: <https://www.newdayoffice.com/blog/how-to-choose-the-best-office-layout-for-your-business>

- C. **Co-Working Office Design.** Co-working office plans are popular with workers without a designated workspace. Desks can be assigned to workers based on their jobs. This arrangement may be ideal for encouraging staff members to interact socially.

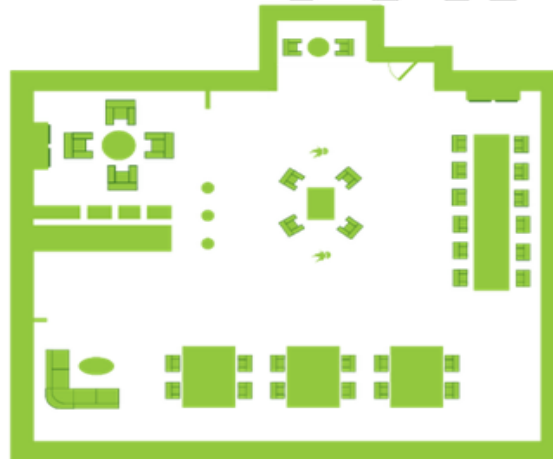


Figure 3: Co-Working Office Design

Source: <https://www.newdayoffice.com/blog/how-to-choose-the-best-office-layout-for-your-business>

- D. **Combination Office Design.** It is a particularly adaptable office layout. A combination office layout might be the best choice to incorporate open and mobile work options. In contrast to the typical cellular layout, you can design individual cubicles that incorporate smaller spaces. You can strategically position office furniture to divide spaces rather than installing windows or doors.

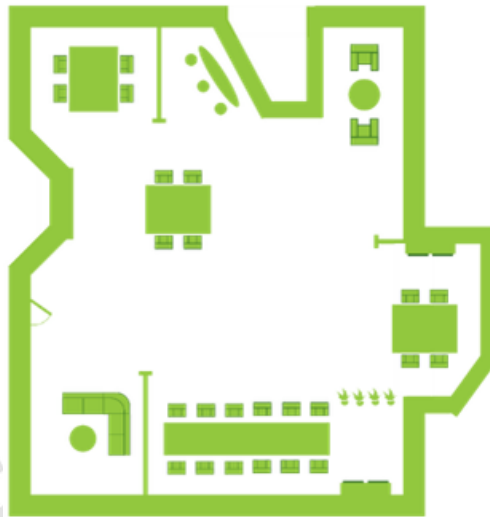


Figure 4: Combination Office Design

Source: <https://www.newdayoffice.com/blog/how-to-choose-the-best-office-layout-for-your-business>

Manufacturing Layout

Manufacturing businesses, such as technopreneurial businesses like Apple, Samsung, Panasonic, Sony, etc., have a range of selections to choose from when designing their production layouts. This depends on the total size of the equipment, products, and employees they use.

Owners must be aware that they must set up their facilities in such a way that it would minimize the travel time of half-finished products between different departments (Ingram, 2016).

In designing the manufacturing layout, a business owner may select four (4) tried and tested options: the product or line layout, functional layout, fixed position layout, or combination layout.

- A. Product or line layout.** In this type of layout, only a single product type is created in an operating area. The product is expected to be standardized and produced in large quantities. Typically, raw materials are supplied at one end of the line and go from one operation to the next until they come out as a finished product. It is commonly used if there is a need to produce more product quantity.

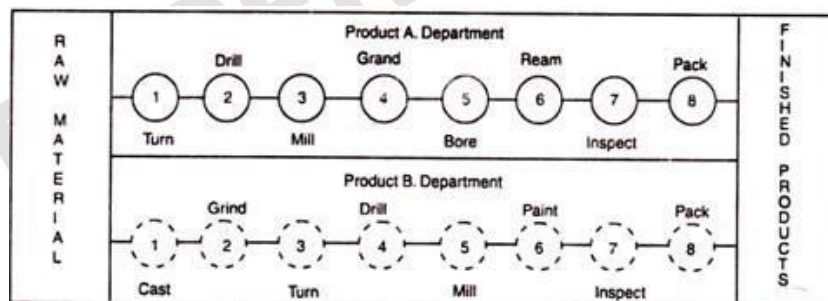


Figure 5. Product or line layout

Source: <http://www.egyankosh.ac.in>

- B. **Process or functional layout.** This layout is helpful if there is a low production need. In this layout, machinery and equipment are positioned according to the nature or type of operations.

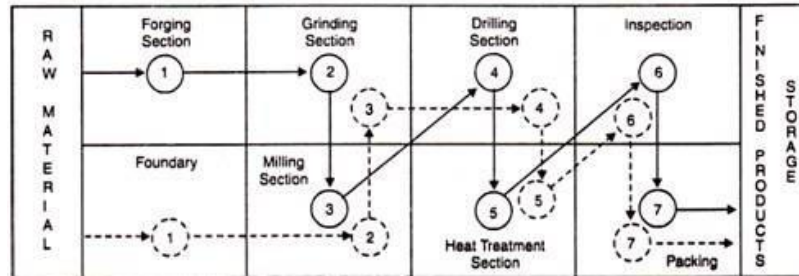


Figure 6. Process or functional layout

Source: <http://www.egyankosh.ac.in>

- C. **Fixed position layout.** In this layout, the major component of a product remains in a fixed position or location. The raw materials, parts, accessories, machinery, and workers are brought to where the products are being built. The product remains fixed because it may be too heavy; the equipment will be permanently installed in a certain place or facility.

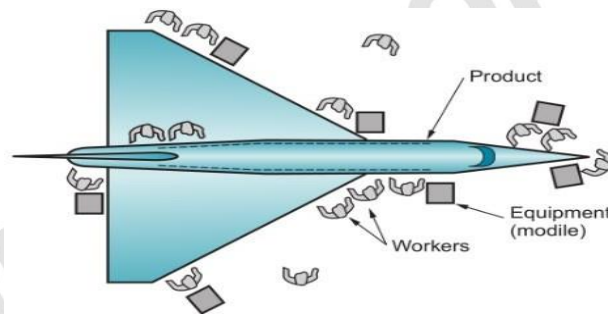


Figure 7. Fixed position layout

Source: slideshare.net

- D. Combination type of layout.** It is a layout that utilizes a combination or a mix of other layouts. This type is possible if an item being made has different types and sizes. In such cases, machinery and equipment are positioned in a process layout. Still, the process grouping (a group of similar machines) is then arranged sequentially to produce various types and sizes of products.

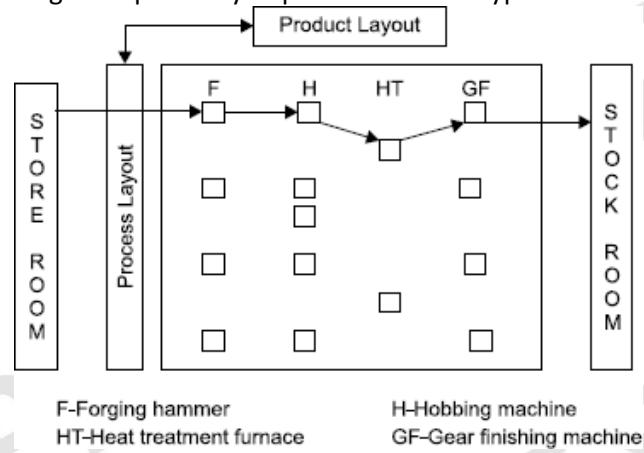


Figure 8. Combination layout
Source: r4cloud.com

Virtual Office

Start-up technopreneurs most commonly use virtual workplaces. It considers an expansion in efficiency, a decrease in driving expenses and greater adaptability. A virtual office has provided flexibility and mobility in business operations. It also made accessibility, sharing, backup, and recovery of data simpler. It enhances collaboration, boosts productivity, but also aids in business continuity.

The steps to setting up a virtual office are as follows (Martin Liam, n.d):

- **Step 1:** Set up a phone system and office address
- **Step 2:** Acquire a domain name and e-mail address
- **Step 3:** Adopt the right tools and equipment
 - Billing and Invoicing Solution
 - File Sharing and Document Management System
 - Backup and Recovery Solution
- **Step 4:** Set up collaboration and communication tools
 - Instant Messaging Apps and Video Conferencing Software
 - Social Media Pages
 - Project Management Tools
- **Step 5:** Establish the organizational rules and regulations

The 5s

The 5s was an idea of Sakichi Toyoda, his son Kiichiro, and a Toyota engineer Taiichi Ohno. 5S became a methodology in all plants of Toyota worldwide after World War II. It was called the Total Production System or the TPS. The three (3) pioneers visited the United States to see how Ford Motor Company

works. While observing its processes, they noticed that time was wasted since workers from the other line needed to wait for one step to be completed before they could work on their share of the production.

The three (3) addressed what they saw and ushered in the 5S methodology. In the 5S method, each employee is important. It also tells them that their work is essential to the process and that what they do contributes to the business's success.

To a layperson, 5S is simply a method of organizing and cleaning the workplace to get the best and most efficient results. However, 5S is a system of organizing workspaces to perform work efficiently, effectively, and safely. The 5S system focuses on putting everything where it belongs and keeping workplaces tidy, thus making it easier for workers to do their jobs and protecting them from possible injury (Sheid, 2018).

The 5S stands for:

- **Seiri** – It means *sorting* in English. This principle tells the employee to organize the important tools and materials they need for their work and to discard the ones they do not need.
- **Seiton** – It means *to set in order* in English. This principle tells the worker to arrange their tools and materials so that few movements will be needed to locate them when they need to use them.
- **Seiso** – It means *cleanliness* or *shine* in English. It emphasizes that cleaning the workplace should be a daily habit.
- **Seiketsu** – It means *standardized* in English. It teaches the employee their role in maintaining and following the first three (3) “Ss” of the 5S system. Part of standardizing is assigning employees regular tasks, schedules, and instructions to make the activities routines.
- **Shitsuke** – It means “*discipline*” or “*sustain*” in English. It teaches the employee that the 5S method is an ongoing process and a continuing responsibility.

Business Model Canvas (Business Management Canvas)

According to an article published by Creately on their website last 2022, a business model canvas, sometimes called a “Business Management Canvas,” is merely a plan that outlines a company's financial goals. It explains who your customer base is, how a business provides value to them, and the specifics of financing that go along with it. Additionally, the business model canvas enables firms to define these parts on a single page.

In its simplest form, a business model canvas provides information about a company's target market, the market's needs, and the role the market offerings will play in satisfying those needs. Businesses will think of ideas and research each aspect as they fill out a Business Model Canvas. They can place the data they collect in any relevant canvas area.

Business Model Canvas (BMC)				
<u>Key Partners:-</u> - Who are the key partners/ Suppliers or vendors? - Do we have any consulting firms? - Which key resources or capabilities do they provide?	<u>Key Activities:-</u> - What are the key processes & tasks we need to create to deliver the value proposition? - What do we develop, sell or manage?	<u>Value Propositions:-</u> - What is the core problem we are trying to solve? - Are there multiple problems we are trying to solve? - How the new value will be different than the current value?	<u>Customer Relationships:-</u> - How we should keep relationship with our customer? - Which idea will fits best with each segments?	<u>Market Segments:-</u> - Who are the primary customer for this product? - For whom we are creating the values? - Do we have any secondary customer? - What are the customer segments or user groups we have identified?
<u>Key Resources:-</u> - Who are the key resources to deliver the key activities? - Do we have any key partners to deliver the same?			<u>Channels:-</u> - How we should connect with your customers? - What are the most cost-efficient and reliable channels?	
<u>Cost Structures:-</u> - What are the fixed & variable costs? - What are the main elements of operational expenses? - Do we have costs involved for CAPEX?			<u>Revenue Structures:-</u> - What is our primary revenue model? - What are the secondary source of revenue? - Do we have subscription, service, licensing fee model?	

Figure 9. Business Model Canvas
Source: <https://miro.medium.com/BMC>

The business model canvas consists of nine (9) components, which are:

1. **Customer Segment** – These are the various groups of people or organizations that an enterprise aims to reach and serve. Any business model is centered on its customers. No business can last very long without (profitable) customers.
2. **Customer Relationship** – Relationships can be personal or automated, transactional or long-term, and the goal can be to get new customers, keep existing ones, or increase sales. The overall customer experience is significantly influenced by the customer relationships you establish.
3. **Channels** – These are ways a company reaches out to specific customer groups. Channels are situated in BMC between Customer Segments and Value Propositions. With this layout, businesses can tailor a particular value to a specific customer segment through the appropriate channel.
4. **Revenue Streams** – Revenue streams are crucial and should align with the business model's cost structure. The profit or loss of the business is the difference between the cost and the revenue streams. In a simple explanation, this is where a business can verify the business model's profitability.
5. **Key Activities** – A business must take these most crucial actions to run smoothly. They are needed to reach markets, maintain customer relationships, create a value proposition, and generate revenue, just like key resources are.
6. **Key Resources** – Key Resources are necessary for each business model. A company can reach markets, maintain relationships with customer segments, and generate revenue thanks to its resources. Depending on the business model, distinct Key Resources are needed.

7. **Key Partners** – Key Partners are relationships a business has with other people or organizations that make the business model work, such as suppliers, manufacturers, or advisors. These partnerships are necessary to succeed in areas where the company alone would be inefficient.
8. **Cost Structure** – The idea of a cost structure is to help determine how to focus on innovation and develop a value proposition. Businesses can aim to cut costs and get the most out of every cost the business incurs by comprehending cost structures.
9. **Value Proposition** - Customers choose a business over others because of its value proposition, which meets their needs or solves the customers' problems. Each Value Proposition is a set of products or services designed to meet the needs of a particular customer segment.

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